CASE STUDY QUESTION

Q5 [Dashboard] Manufacturing Operations

- Design a Manufacturing Operations Dashboard with:
  - KPI tiles — Total Units Produced, Defect Rate %, Downtime Hours, and Scrap Cost,
  - A Bar chart for Units Produced by Plant,
  - A Line chart for Monthly Defect Trend,
  - A Pie chart for Defect Type Distribution.
  - A table for Top 5 Suppliers by Material Cost.

  Add Plant and Date Range filters.

CASE STUDY
Manufacturing Operations Dashboard

Create a comprehensive Manufacturing Operations Dashboard with key visualizations for real-time operational monitoring. Include **KPI tiles**, charts, and a supplier table with filters for Plant and Date Range selection.

Steps to Create the Dashboard:

54,320	3.8%	127	$38,250	$38,250
TOTAL UNITS PRODUCED	DEFECT RATE %	DOWNTIME HOURS	SCRAP COST	SCRAP COST

01 Add KPI Tiles

Add **KPI tiles** at the top with

- Total Units Produced, **Defect Rate %**,
- Downtime Hours, **Scrap Cost.**

02 Create Plant Units Bar Chart.

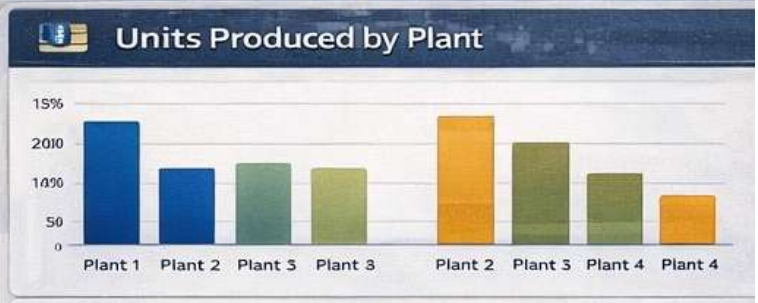
Design a Bar Chart showing Units Produced grouped by Plant.

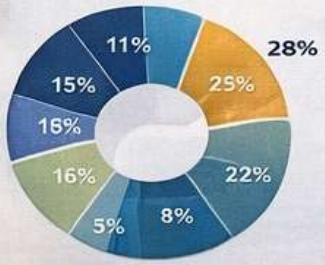
03 Add Defect Trend & Distribution Charts

- Include a Line Chart for Monthly **Defect Trend.**
- Add a Pie Chart for Defect Type Distribution

04 Insert Top Supplier Table

Add a table showing the Top 5 Suppliers by Material Cost.





Defect Type Distribution

Scratch	23%
Misalignment	21%
Burr	18%
Dent	16%
Discoloration	10%
Weld Spatter	8%

Edit 100% Analytics Filters

Tab 1

Total Units Produced

4.19K

Units Produced

Defect Rate %

267.11

Defect Rate %

Downtime Hours

102.5

Downtime Hours

Scrap Cost

7.84M

Scrap Cost

Plant Location

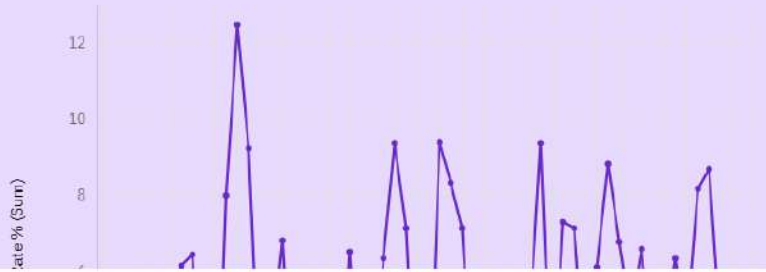
Chennai

Production Date

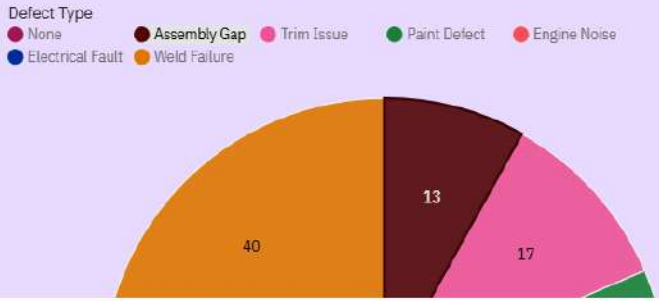
Select value

Units Produced by Plant

Monthly Defect Trend



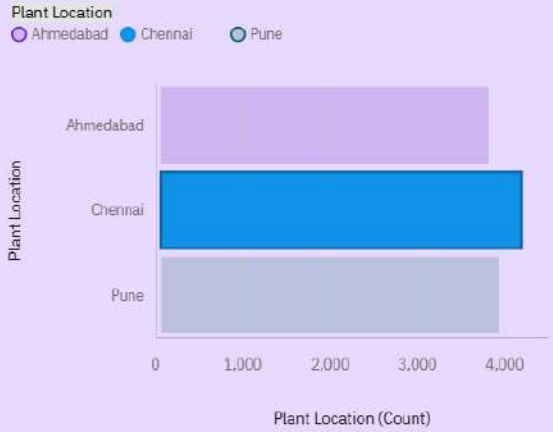
Defect Type Distribution



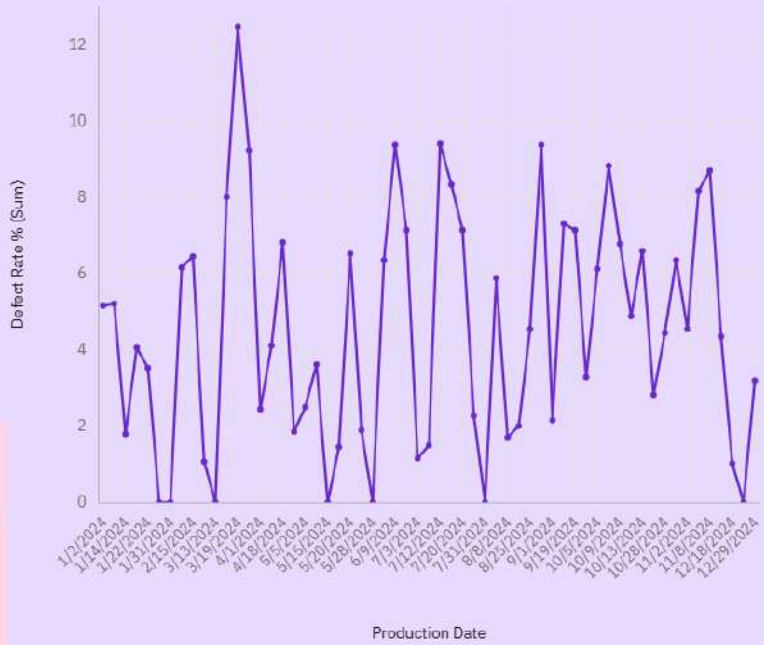
Edit 100% Analytics Filters

Tab 1

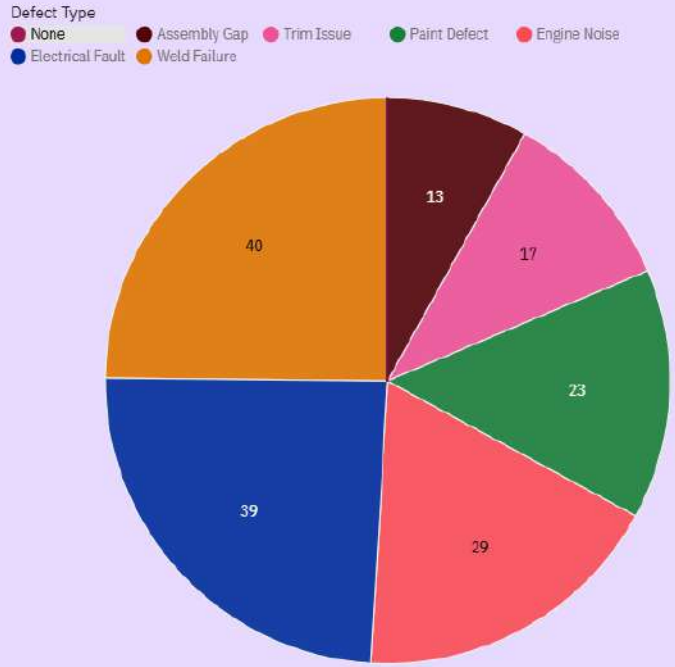
Units Produced by Plant



Monthly Defect Trend

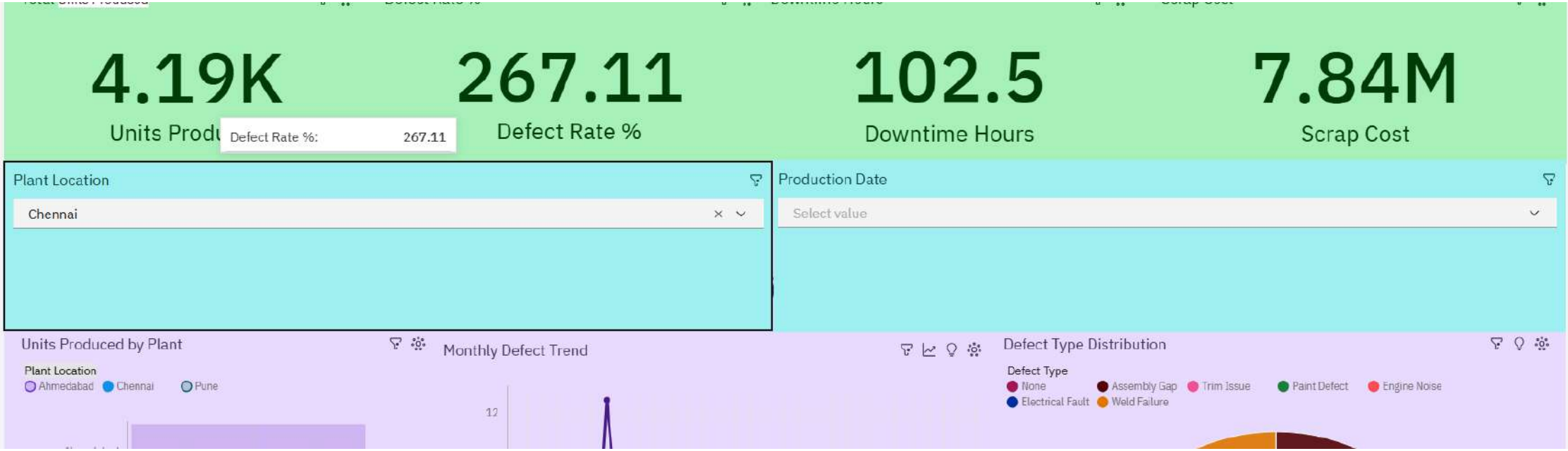


Defect Type Distribution



Raw Material Supplier and Material Cost per Unit

Rank (Material Cost p...	Raw Material Supplier	Material Cost per Unit
1	GlassMart	2,016,033.13
2	AutoParts India	2,001,256.78
3	ElecComp	1,712,820.22


4.19K
Units Produ
Defect Rate %: 267.11
267.11
Defect Rate %
102.5
Downtime Hours
7.84M
Scrap Cost
Plant Location
Chennai
Production Date
Select value
Units Produced by Plant
Plant Location
Ahmedabad
Chennai
Pune
Monthly Defect Trend
12
Defect Type Distribution
Defect Type
None
Assembly Gap
Trim Issue
Paint Defect
Engine Noise
Electrical Fault
Weld Failure

Tab 1

Total Units Produced

4.19K

Units Produced

Defect Rate %

267.11

Defect Rate %

Downtime Hours

102.5

Downtime Hours

Scrap Cost

7.84M

Scrap Cost

Plant Location

Chennai

Production Date

Select value

2024-01-02

2024-01-13

2024-01-14

2024-01-15

2024-01-22

Units Produced by Plant

Plant Location

Ahmedabad Chennai Pune

Ahmedabad

Chennai

Monthly Defect Trend

12

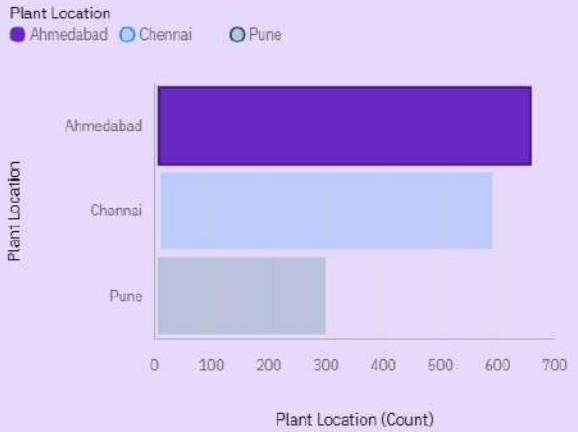
10

8

Electrical Fault Weld Failure

13

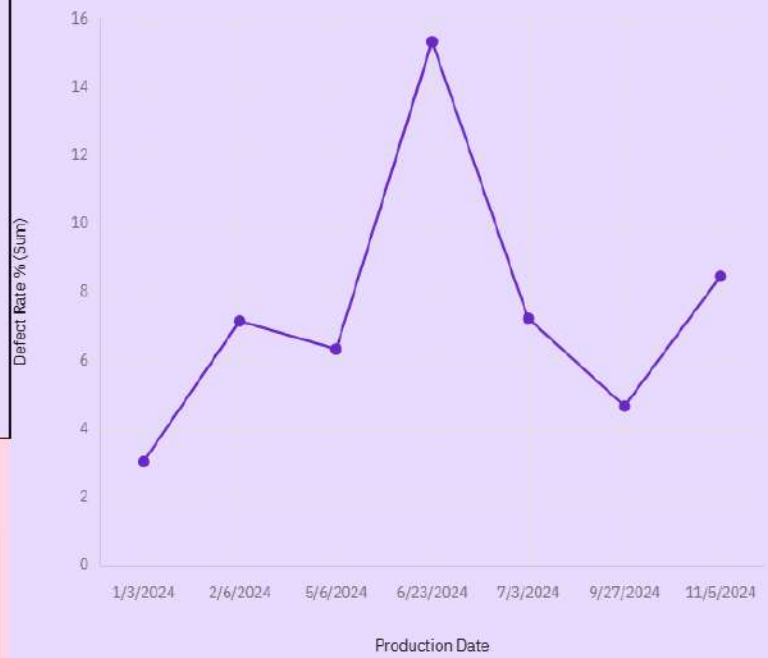
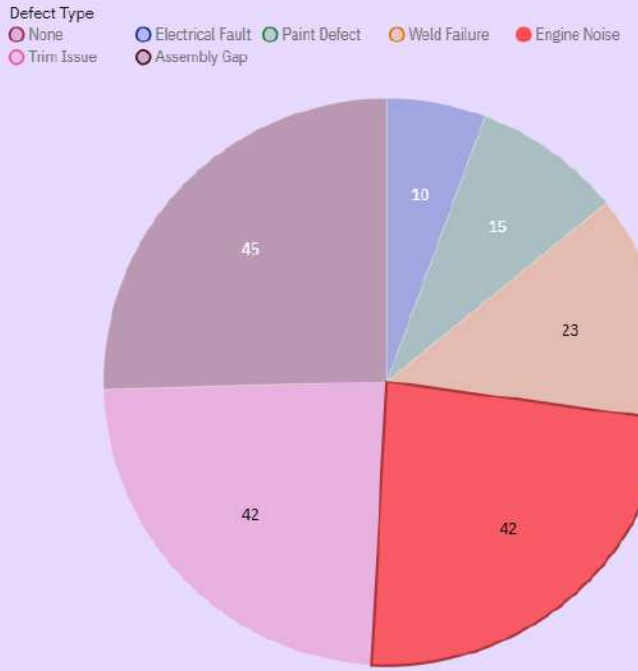
Units Produced by Plant



Raw Material Supplier and Material Cost per Unit

Rank (Material Cost pe...	Raw Material Supplier	Material Cost per Unit
1	PlastiTech	406,045.0
2	SteelCo	367,579.1
3	GlassMart	164,918.7

Monthly Defect Trend



Defect Type Distribution